

Conditional Probability

1. Suppose that there are two different diagnostic tests (say, Test A and Test B) for a particular disease of interest. In a certain large population, suppose that the prevalence of this disease is 1%. Among all those people who have this disease in this population, 10% will incorrectly test negatively for the presence of the disease when given Test A; and, independently of any results based on Test A, 5% of these diseased people will incorrectly test negatively when given Test B. Among all those people who do *not* have the disease in this population, 6% will incorrectly test positively when given Test A; and, independently of any results based on Test A, 8% of these non-diseased people will incorrectly test positively when given Test B. (Note that the results of tests A and B are *conditionally* independent. That is, conditioned on disease status, tests A and B are independent, but they are not necessarily *unconditionally* independent.)
 - (a) Given that both Tests A and B are positive when administered to a person selected randomly from this population, what is the probability that this person actually has the disease?
 - (b) Given that Test A is positive when administered to a person selected randomly from this population, what is the probability that Test B will also be positive?
 - (c) Given that a person selected randomly from this population actually has the disease, what is the probability that at least one of the tests is positive?

Independence

2. CB 1.38
3. CB 1.39
4. CB 1.40
5. Show that if A , B , and C are mutually independent, then $A \cap B$ and C are independent and $A \cup B$ and C are independent.

Random Variables and Distributions

6. CB 1.47 (parts b, d and e)
7. CB 1.53
8. Consider a random variable, X , with pdf

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $F(x)$ and prove it is a valid cdf.

9. Consider a random variable X with pdf,

$$f_X(x) = \begin{cases} \frac{x}{4} & 0 < x \leq 2, \\ e^{-2(x-2)} & x > 2 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the CDF of X and prove that it is valid.
- (b) Is X a continuous random variable? Why or why not?

10. Consider a random variable, X , with cdf

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x+1}{2} & x \in [0, 1) \\ 1 & x \geq 1 \end{cases}$$

Draw a picture of the cdf. Is this random variable continuous, discrete, or neither. Why?

11. In each of the following find the value of c so that $f(x)$ is a valid pmf or pdf:

(a)

$$f(x) = P(X = x) = \frac{c}{2^x} \quad \text{for} \quad x = 0, 1, 2, \dots$$

(b)

$$f(x) = cx^2 \quad \text{for} \quad 0 \leq x \leq 1$$

12. Suppose that X has the density function $f(x) = cx^2$ for $0 \leq x \leq 1$ and $f(x) = 0$ otherwise.

(a) Find c .

(b) Find the cdf.

(c) What is $P(.1 \leq X \leq .5)$?

13. Suppose that the lifetime of an electronic component follows an exponential distribution with $\lambda = .1$.

(a) Find the probability that the lifetime is less than 10.

(b) Find the probability that the lifetime is between 5 and 15.

(c) Find t such that the probability that the lifetime is greater than t is .01.

14. T is an exponential random variable, and $P(T < 1) = .05$. What is λ ?